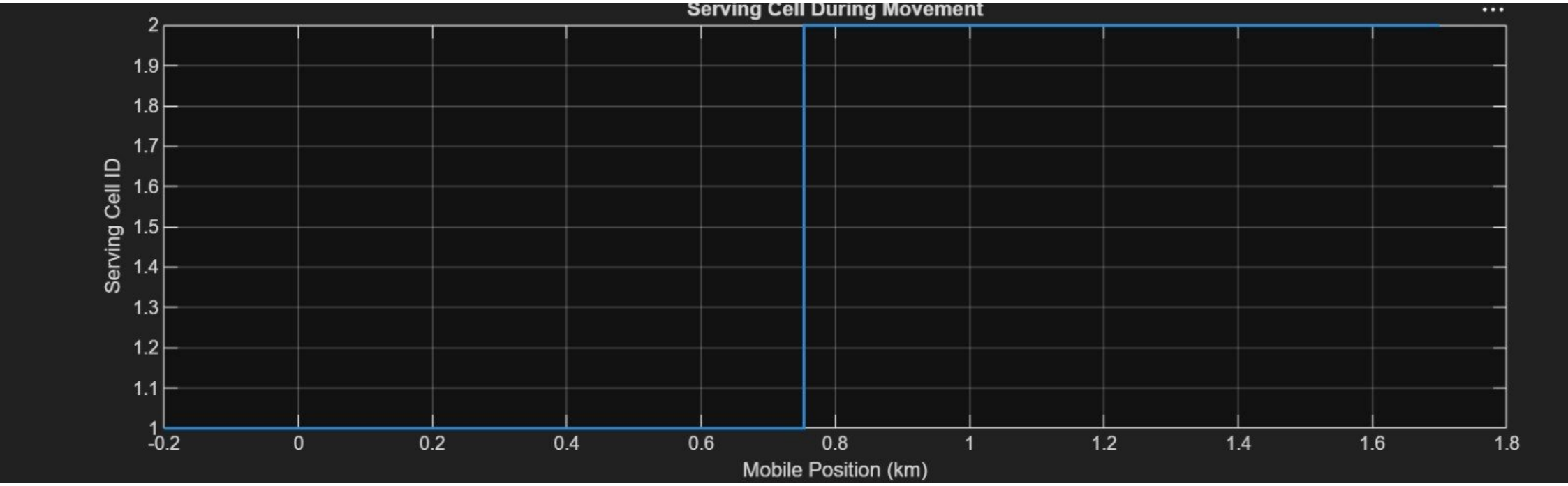
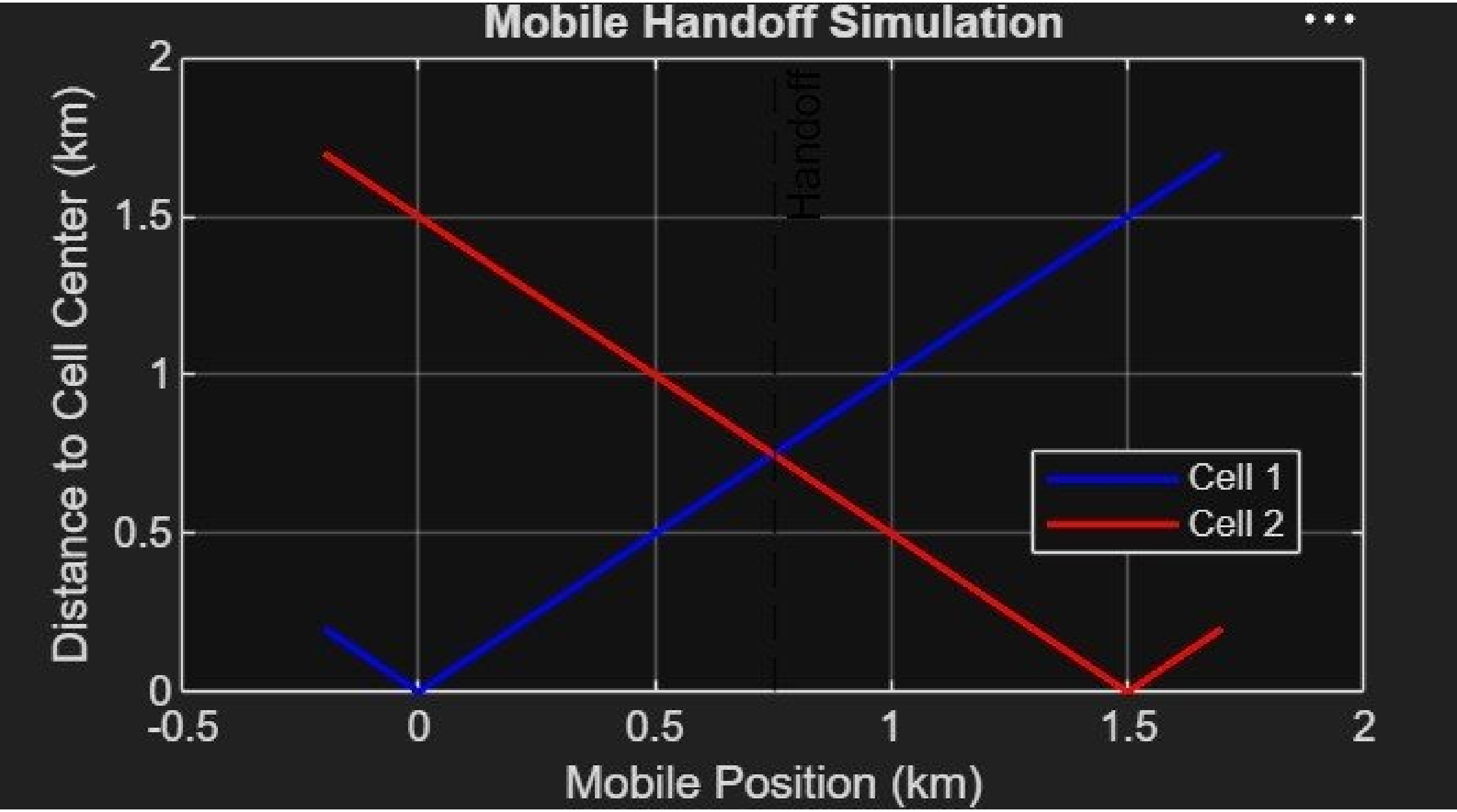
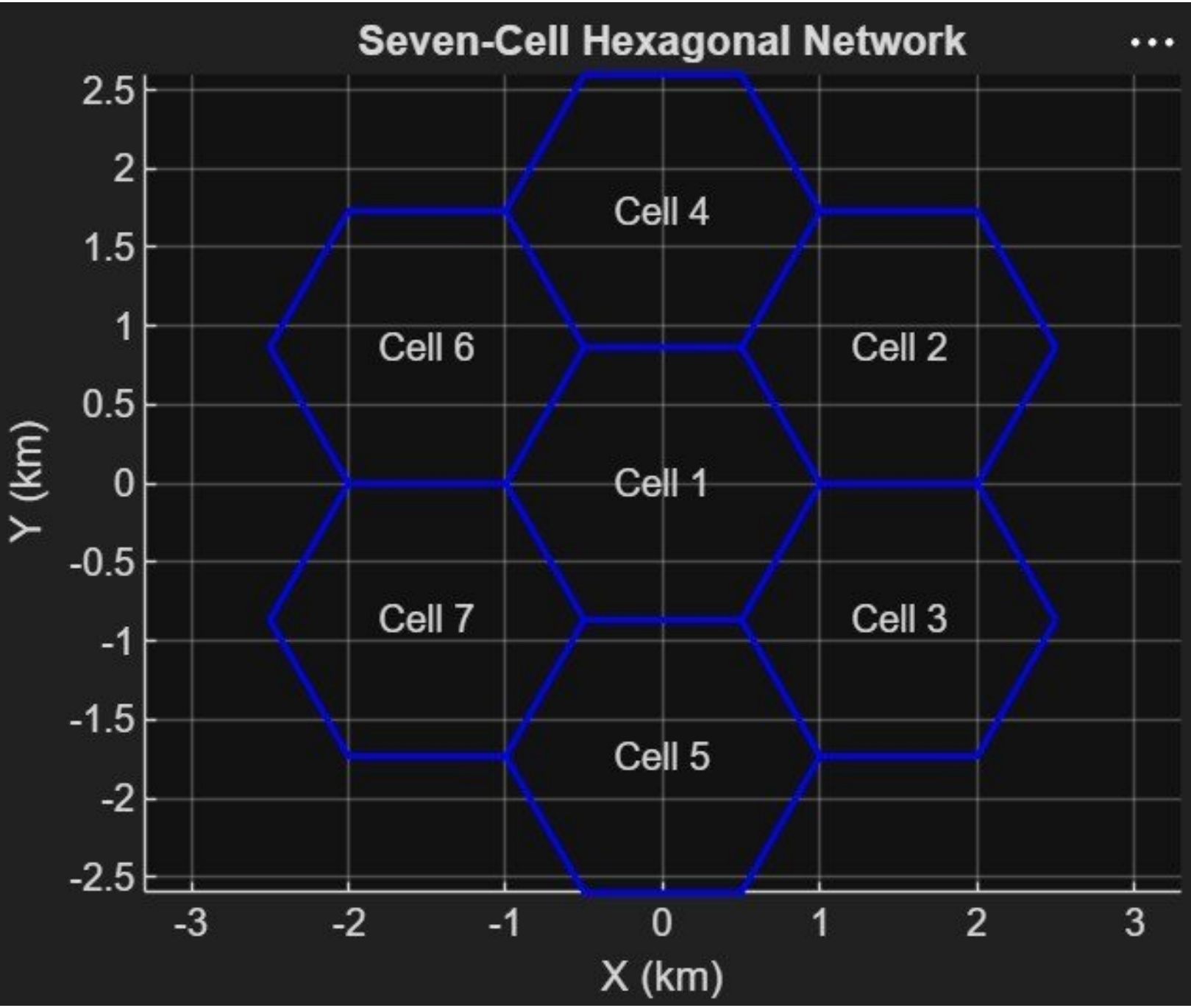
```
clear;
clc;
close all;
% Parameters
serviceAreaKm2 = 5;
radiiKm = [1.0 0.5];
usersPerCell = 30;
% Calculate cell area and capacity
hexArea = @(R) (3*sqrt(3)/2).*R.^2;
cellAreaKm2 = hexArea(radiiKm);
cellCount = ceil(serviceAreaKm2 ./ cellAreaKm2);
estimatedUsers = cellCount .* usersPerCell;
% Display results
T = table(radiiKm', cellAreaKm2', cellCount', estimatedUsers', ...
    'VariableNames', {'Radius_km','CellArea_km2', ...
    'EstimatedCells','TotalUsers'});
disp(T);
% Draw seven hexagonal cells
R = radiiKm(1);
centers = [0 0;
    1.5*R sqrt(3)*R/2;
    1.5*R -sqrt(3)*R/2;
    0 sqrt(3)*R;
    0 -sqrt(3)*R;
    -1.5*R sqrt(3)*R/2;
    -1.5*R -sqrt(3)*R/2];
theta = (0:6)*pi/3;
figure;
hold on;
axis equal;
```

```
grid on;
for k = 1:size(centers,1)
    x = centers(k,1) + R*cos(theta);
    y = centers(k,2) + R*sin(theta);
    plot(x,y,'b-','LineWidth',1.5);
    text(centers(k,1),centers(k,2), ...
        sprintf('Cell %d',k), ...
        'HorizontalAlignment','center');
end
xlabel('X (km)');
ylabel('Y (km)');
title('Seven-Cell Hexagonal Network');
hold off;
% Simulate mobile movement and handoff
mobileX = linspace(-0.2*R,1.7*R,300);
mobileY = zeros(size(mobileX));
d1 = hypot(mobileX,mobileY);
d2 = hypot(mobileX-1.5*R,mobileY);
servingCell = ones(size(mobileX));
servingCell(d2 < d1) = 2;
handoffIndex = find(diff(servingCell)~=0,1) + 1;
figure;
plot(mobileX,d1,'b','LineWidth',1.5);
hold on;
plot(mobileX,d2,'r','LineWidth',1.5);
if ~isempty(handoffIndex)
    xline(mobileX(handoffIndex),'--k','Handoff');
end
xlabel('Mobile Position (km)');
ylabel('Distance to Cell Center (km)');
```

```
ylabel('Y (km)');
title('Seven-Cell Hexagonal Network');
hold off;
% Simulate mobile movement and handoff
mobileX = linspace(-0.2*R,1.7*R,300);
mobileY = zeros(size(mobileX));
d1 = hypot(mobileX,mobileY);
d2 = hypot(mobileX-1.5*R,mobileY);
servingCell = ones(size(mobileX));
servingCell(d2 < d1) = 2;
handoffIndex = find(diff(servingCell)~=0,1) + 1;
figure;
plot(mobileX,d1,'b','LineWidth',1.5);
hold on;
plot(mobileX,d2,'r','LineWidth',1.5);
if ~isempty(handoffIndex)
    xline(mobileX(handoffIndex),'--k','Handoff');
end
xlabel('Mobile Position (km)');
ylabel('Distance to Cell Center (km)');
title('Mobile Handoff Simulation');
legend('Cell 1','Cell 2','Location','best');
grid on;
hold off;
figure;
stairs(mobileX,servingCell,'LineWidth',1.5);
xlabel('Mobile Position (km)');
ylabel('Serving Cell ID');
title('Serving Cell During Movement');
grid on;
```


Serving Cell During Movement
Serving Cell ID
2
1.9
1.8
1.7
1.6
1.5
1.4
1.3
1.2
1.1
1
-0.2
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
Mobile Position (km)


Mobile Handoff Simulation
Distance to Cell Center (km)
Mobile Position (km)
Handoff
Cell 1
Cell 2
2
1.5
1
0.5
0
-0.5
0
0.5
1
1.5
2


Seven-Cell Hexagonal Network
Cell 4
Cell 6
Cell 2
Cell 1
Cell 7
Cell 3
Cell 5
2.5
2
1.5
1
0.5
0
-0.5
-1
-1.5
-2
-2.5
-3
-2
-1
0
1
2
3
Y (km)
X (km)

Command Window

Radius_km	CellArea_km2	EstimatedCells	TotalUsers
1	2.5981	2	60
0.5	0.64952	8	240

>>